Tutorial 3.1

Question 4

Ahmad and Syahmi belong to an athletics club. In javelin throwing competitions, Ahmad throws is normally distributed with mean 41.0 m and standard deviation 2.0 m. While, 85% of Syahmi's javelin throws exceed 35m and 70% exceed 37.5 m. Syahmi's throws is also normally distributed, with standard deviation, $\sigma = 6.75$.

a) What is the probability of Ahmad's throws between 40 m and 46 m? (2 marks)

b) Find the mean of Syahmi's throws, in two decimal places. (3 marks)

c) In order to qualify for the final round, it is necessary to achieve a throw of at least 48 m in the preliminary rounds. (3 marks)

I. Calculate the probability of Ahmad qualify for the final round.

II. Calculate the probability of Syahmi qualify for the final round.

d) Based on results in (c), who should be chosen, Ahmad or Syahmi? Justify your answer. (1 mark)

Answer :

Ahmad:

$$X_A \sim N(41,2)$$

Syahmi:

$$X_S \sim N(\mu, 6.75)$$

a) Probability Ahmad throws between 40 m and 46 m

Find Z-values:

$$Z_1 = \frac{40 - 41}{2} = -0.50$$
$$Z_2 = \frac{46 - 41}{2} = 2.50$$

So:

$$P(40 < X < 46) = P(-0.50 < Z < 2.50)$$

From Z-table:

$$P(Z < 2.50) = 0.9938$$
$$P(Z < -0.50) = 0.3085$$

Therefore:

$$P(40 < X < 46) = 0.9938 - 0.3085 = 0.6853$$

Ahmad has about **68.53%** probability of throwing between 40 m and 46 m.

b) Find Syahmi's mean

Given:

$$70\% \text{ exceed } 37.5$$

So:

$$P(X > 37.5) = 0.70$$

Therefore:

$$P(X < 37.5) = 0.30$$

From Z-table:

$$P(Z < -0.52) = 0.30$$

So:

$$Z = -0.52$$

Use:

$$Z = \frac{X - \mu}{\sigma}$$

Substitute values:

$$-0.52 = \frac{37.5 - \mu}{6.75}$$
$$-0.52(6.75) = 37.5 - \mu$$
$$-3.51 = 37.5 - \mu$$
$$\mu = 37.5 + 3.51$$
$$\mu = 41.01$$

c)(I) Probability Ahmad qualifies: at least 48 m

Need:

$$P(X \geq 48)$$

For Ahmad:

$$Z = \frac{48 - 41}{2} = 3.50$$

From Z-table:

$$P(Z < 3.50) = 0.9998$$

Therefore:

$$P(Z \geq 3.50) = 1 - 0.9998 = 0.0002$$

c)(II) Probability Syahmi qualifies: at least 48 m

Using:

$$\mu = 41.01, \sigma = 6.75$$
$$Z = \frac{48 - 41.01}{6.75}$$
$$Z = \frac{6.99}{6.75} = 1.04$$

From Z-table:

$$P(Z < 1.04) = 0.8508$$

Therefore:

$$P(Z \geq 1.04) = 1 - 0.8508 = 0.1492$$

d) Who should be chosen?

Syahmi should be chosen because his probability of qualifying is higher.

Ahmad:

$$0.0002 = 0.02\%$$

Syahmi:

$$0.1492 = 14.92\%$$

This is because Syahmi has a much higher probability of achieving at least 48 m.